

- 3 The Earth may be assumed to be a uniform sphere of radius R and mass M . At its surface, the gravitational field strength is g . A satellite orbits the Earth at a height $0.30R$ above its surface.

(a) Show that the gravitational field strength at this height is $0.59g$.

$$g = \frac{GM}{r^2} \Rightarrow g \propto \frac{1}{r^2}$$

$$\frac{g'}{g} = \frac{r^2}{(r')^2} \quad [\text{B1}]$$

$$g' = \frac{R^2}{(1.3R)^2} g \quad [\text{B1}]$$

$$g' = 0.59g \quad (\text{shown})$$

[2]

- (b) Determine the angular speed of the satellite about the Earth. The radius R of the Earth is 6.4×10^6 m.

The gravitational force by Earth provides the centripetal force on the mass.

$$mg = ma_c$$

$$0.59g = 1.3(6.4 \times 10^6)\omega^2 \quad [\text{B1}]$$

$$\omega = 8.3 \times 10^{-4} \text{ rad s}^{-1} \quad [\text{B1}]$$

angular speed = rad s⁻¹ [2]

- (c) Calculate the time, in hours, for one complete orbit of the satellite.

$$\omega = \frac{2\pi}{T}$$

$$8.34 \times 10^{-4} = \frac{2\pi}{T}$$

$$T = 7.5 \times 10^3 \text{ s} \quad [\text{B1}]$$

$$T = 2.1 \text{ h} \quad [\text{B1}]$$

time = h [2]

- (d) Explain why the satellite does not fall towards the Earth even though the gravitational force is directed toward the centre of the Earth.

The satellite is rotating about the Earth. The force exerted by Earth on the satellite is

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perpendicular to the linear velocity of the satellite [B1], and the magnitude of this force

.....
provides the centripetal acceleration required for the satellite to revolve around the Earth

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[B1] at a constant distance and a constant speed. $F = \frac{mv^2}{r}$ [2]

[Total: 8]

4. The piston in the cylinder of a car engine is made to move in the cylinder with simple harmonic